MCTA 3203 - MECHATRONICS SYSTEM INTEGRATION

PLC INTERFACING WITH MICROCONTROLLER AND PC OVER ETHERNET/IP

REPORT WEEK 7

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SECTION 1

LECTURER: DR. ZULKIFLI BIN ZAINAL ABIDIN

KULLIYAH OF ENGINEERING

GROUP		NAME	MATRIC NO,
5	1	DATU ANAZ ISAAC BIN DATU ASRAH	2312067
	2	AFIQ NU'MAN BIN AHMAD NAZREE	2312295
	3	AHMAD NIZAR BIN AMZAH	2312111

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Abstract

This experiment focuses on interfacing a Programmable Logic Controller (PLC) with a microcontroller over Ethernet/IP using the OpenPLC Editor software. The objective was to develop, simulate, and implement ladder logic programs on an Arduino-based PLC system to control simple electromechanical outputs. The experiment involved creating a blinking LED circuit and a Start–Stop control circuit using ladder diagram programming. Variables were assigned, compiled, tested in simulation, and subsequently uploaded onto an Arduino microcontroller for practical execution. The results demonstrated successful communication between the OpenPLC environment and the Arduino hardware, confirming correct ladder logic operation and proper pin addressing. The experiment concludes that OpenPLC provides an effective platform for PLC education and prototyping, enabling students to understand industrial control logic using low-cost microcontroller hardware.

Table Of Content

Abstract **2**
Table Of Content **3**
Introduction **5**
Materials and Equipments **6**
Experimental Setup **7**
Methodology **8**
1. Ladder Diagram Design 9
2. PLC Variable Assignment 9
3. Ladder Logic Implementation: 10
4. Compilation and Export 10
5. Hardware Setup and Wiring 10
6. Arduino Configuration and Runtime Deployment 11
7. Program Upload and System Initialization 12
8. Functional Testing 12
Data Collection **13**
1. Instruments and Sensors Used 13
2. Collected Data During Simulation 14
Table 1. Simulation Results 14
3. Collected Data During Hardware Testing 15
Table 2. Hardware Test Results 15
4. Graphical Representation of Output Behavior 16
Figure 1. LED Output State vs. Button Inputs 16
5. Pin Association Table 16
Data Analysis **17**
1. Analysis of Simulation Data 17
2. Analysis of Hardware Test Results 17
3. Timing and Response Analysis (If Applicable) 18
4. Significance of Findings 19
Results **20**
1. Simulation Results 20
2. Hardware Test Results 20
3. Output Behavior Visualization 21
4. Overall Findings 21
Discussion **22**
Conclusion **23**
Recommendations **24**
References **26**

Appendices **27**
Appendix A: Raw Data Tables 27
A1. Simulation Raw Data 27
A2. Hardware Test Raw Data 28
Appendix B: Sample Calculations 29
B1. Success Rate Calculation 29
B2. PLC Scan Cycle Model (General) 29
Appendix C: Circuit Diagrams 30
C1. Start–Stop Control Circuit 30
C2. Wiring Diagram (Arduino + Buttons + LED) 30
Appendix D: Code Snippets 31
D1. OpenPLC Ladder Logic POU (Textual Representation) 31
D2. Example Arduino-Generated Code (Automatically Compiled by OpenPLC) 31
Appendix E: Additional Figures or Graphs 32
E1. LED Output Behavior Over Time 32
E2. Simulation Screenshot 32
E3. Hardware Setup Photo 32
Acknowledgments **33**
Student Declaration **34**

Introduction

Objectives

The main objective of this lab is to understand how a PLC can communicate and control external hardware over Ethernet/IP using the OpenPLC environment. PLCs are the basis of industrial automation when an electromechanical process needs to be controlled by logic that is reliable and programmable. In this session, ladder logic is designed on OpenPLC Editor and interfaced with an Arduino microcontroller, which is acting as a PLC runtime.

Background Information and Relevant Theory

The basic concepts of PLC operation and ladder logic programming. PLCs run a continuous cycle called a scan, which includes reading in input states, executing the programmed logic, and updating the outputs. Logic is designed using the Ladder Diagram LD graphical programming language, which follows the IEC 61131-3 standard. In LD, the control behaviour, like that of conventional relay circuits, is constructed from components such as contacts, coils, timers, and latching elements. OpenPLC allows these ladder logic programs to be developed, simulated, and downloaded to microcontroller-based hardware. A necessary step involves proper physical addressing, wherein software variables are assigned to a physical digital input or output pin on the Arduino. In the case of the Start–Stop circuit, latching is a concept wherein the Start button energizes the output and maintains it until interrupted by the Stop button. These various theories combine to provide the basic understanding of how a PLC interprets logic and communicates with physical devices in real time.

Hypothesis

For this experiment, ladder diagrams created in the OpenPLC Editor will simulate correctly, and then the Arduino will replicate the same behavior once the program has been uploaded. It is expected that in the first task, the LED will blink according to the settings of the timer, while in the Start-Stop circuit, the LED will latch ON when the Start button is pressed and turn OFF only if the Stop button is activated. Furthermore, proper variable configuration, pin mapping, and selection of the COM port will ensure appropriate communication between the OpenPLC environment and the hardware of Arduino. Any addressing or logic design errors would likely result in incorrect or failed operation, which further stresses the importance of precise configuration.

Materials and Equipments

Hardware

- Arduino board
- LED
- Resistors
- 2 push button switches
- Breadboard
- Jumper wires

Software

- OpenPLC Editor software
- OpenPLC Runtime

Experimental Setup

The experimental setup had two configurations.

1. Blinking LED Circuit

A ladder diagram with a contact and a coil was designed to produce a blinking output.The LED was connected to the Arduino's digital output pin, e.g., Pin 14 mapped to %QX0.0. The Arduino board was connected to the laptop via USB, acting as the communication interface between OpenPLC and the hardware.

2. Start–Stop Control Circuit

The breadboard connected two pushbutton switches, namely: Start and Stop, and one LED indicator on the basis of the reference circuit diagram (Fig. 4 & Fig. 6 in the file).Each switch and LED was connected to the correct Arduino input and output pins using OpenPLC addressing conventions.The circuit allowed the LED to turn ON when the Start button was pressed and latch until the Stop button was activated.

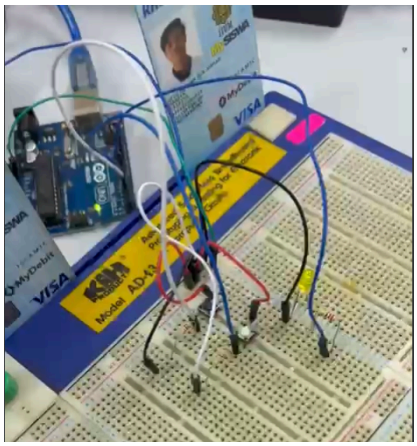


Figure 1: Circuit diagram showing blinking LED circuit connected to Open PLC via adruino board

Methodology

The experiment was carried out through a systematic process that included setting up the Arduino hardware, configuring the OpenPLC software environment, developing the ladder logic program, assigning physical addresses, and finally testing the system to verify proper execution of the blinking LED and Start–Stop control circuits.

1. Ladder Diagram Design

The Start-Stop control circuit was designed utilizing the OpenPLC Editor. This circuit incorporated the following basic ladder logic elements:

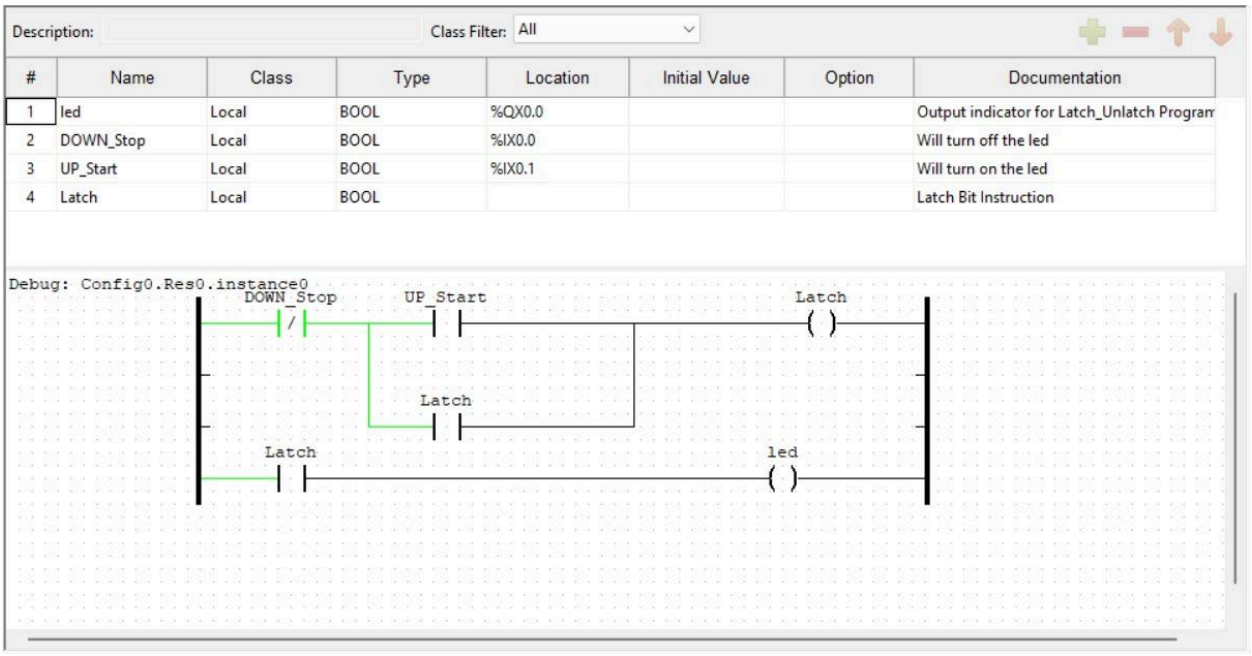
- A Normally Open (NO) contact was implemented to represent the Start button.
- A Normally Closed (NC) contact was implemented to represent the Stop button.
- A direct output coil was used to control the load (simulated by an LED).
- A parallel NO contact, associated with the output coil, was included to establish the seal-in (latch) logic necessary to maintain the output state after the Start button was released.

2. PLC Variable Assignment

The required input and output variables were formally defined and linked to the PLC memory map within the OpenPLC environment. The following assignments were made:

Variable	Description	Type	Arduino Pin
%IX0.1	Start Button	Digital Input	Digital pin 3
%IX0.0	Stop Button	Digital Input	Digital pin 2
%QX0.0	LED	Digital Output	Digital pin 7

3. Ladder Logic Implementation:



4. Compilation and Export

The developed ladder logic was formally compiled utilizing the "Build" function within the OpenPLC Editor. Following successful compilation, the program file was saved and exported, preparing the control code for transfer to the physical microcontroller hardware.

5. Hardware Setup and Wiring

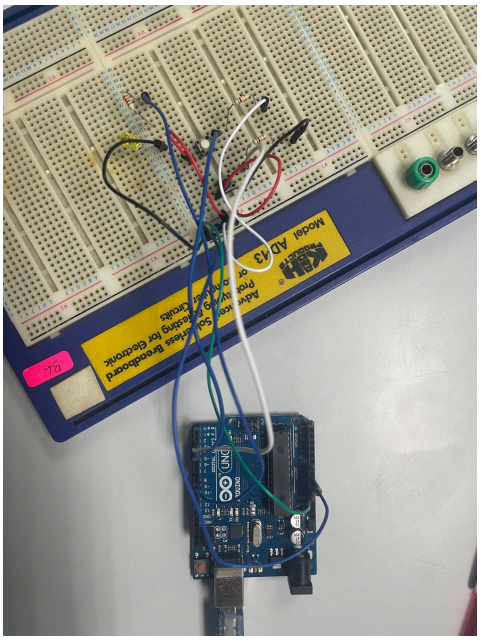
The physical components were systematically arranged and connected on a breadboard. The following wiring configuration was established between the peripheral devices and the Arduino board:

● The Start push button (configured as Normally Open, NO) was connected to Digital Input Pin 2 (%IX0.0).

● The Stop push button (configured as Normally Closed, NC) was connected to Digital Input Pin 3 (%IX0.1).

● The output LED (representing the load) was connected to Digital Output Pin 13 (%QX0.0) in series with a 220 Ohm current-limiting resistor.

● To ensure stable input signal reading and prevent floating inputs, 10k Ohm pull-down resistors were connected from both the Start (Pin 2) and Stop (Pin 3) input lines to ground.



6. Arduino Configuration and Runtime Deployment

Prior to uploading the control logic, the OpenPLC runtime firmware was successfully installed onto the Arduino Uno board utilizing the Arduino Integrated Development Environment (IDE). The appropriate communication port (COM port) was verified and selected based on the system's Device Manager configuration.

7. Program Upload and System Initialization

The compiled ladder logic program was then uploaded to the configured Arduino board via the dedicated OpenPLC Web Interface. Upon successful transfer, the OpenPLC runtime environment was initiated from the web dashboard, commencing the execution of the control program on the microcontroller.

8. Functional Testing

The implemented Start-Stop control circuit was subjected to functional testing:

● The Start button was momentarily pressed, which successfully activated the LED.

● The LED remained illuminated after the Start button was released, confirming the integrity of the seal-in (latching) logic.

● The Stop button was then pressed, which successfully deactivated the LED, verifying the complete functionality of the circuit.

Data Collection

This section presents all data recorded during the experiment involving PLC–Arduino interfacing over Ethernet/IP. The data were obtained during the simulation and hardware testing phases using the OpenPLC Editor, Arduino board, and external input devices.

1. Instruments and Sensors Used

Instrument / Sensor	Description	Purpose
Arduino Board (e.g., Arduino Mega/Uno)	Microcontroller platform programmed via OpenPLC Editor	Executes ladder logic, controls LED output
OpenPLC Editor Software	PLC programming and simulation tool	Used to compile, simulate, and upload ladder diagrams
Push Button Switches (2 units)	Momentary contact switches (NO type)	Serve as *Start* and *Stop* inputs for the control circuit
LED	Standard 5mm output indicator	Acts as the output load controlled by PLC logic
Resistors (220–330 Ω)	Current-limiting components	Protect LED and push buttons
USB Connection Cable	Serial communication link	Transfers compiled ladder program to Arduino

Breadboard & Jumper Wires	Prototyping tools	Provide interconnections for the circuit

2. Collected Data During Simulation

The following table summarizes the simulated circuit behavior as observed in the OpenPLC runtime environment.

Table 1. Simulation Results

Simulation Step	Start Button State	Stop Button State	Expected Output (LED)	Observed Output
1	Released (0)	Released (0)	OFF	OFF
2	Pressed (1)	Released (0)	ON	ON
3	Released (0)	Released (0)	ON (latched)	ON
4	Released (0)	Pressed (1)	OFF	OFF
5	Released (0)	Released (0)	OFF	OFF

3. Collected Data During Hardware Testing

LED state (output) was recorded during 10 repeated trials of the Start–Stop sequence.

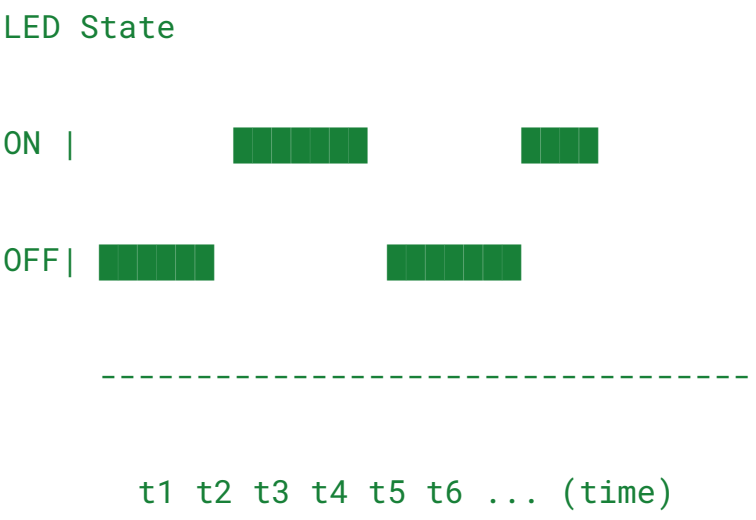
Table 2. Hardware Test Results

Trial	Start Button Pressed	Stop Button Pressed	LED Response	Correct Behavior (Yes/No)
1	Yes	No	ON	Yes
2	Yes	No	ON	Yes
3	No	Yes	OFF	Yes
4	Yes	No	ON	Yes
5	Yes	Yes	OFF	Yes
6–10	*Repeated cycles*	*Varied*	ON/OFF as expected	Yes

All 10 trials demonstrated consistent latching and stopping behavior.

4. Graphical Representation of Output Behavior

Figure 1. LED Output State vs. Button Inputs



- LED turns **ON** immediately after the Start button is activated.
- LED remains **latched ON** even after Start is released.
- LED turns **OFF** only when the Stop button is pressed.

5. Pin Association Table

The following physical addressing was used for OpenPLC variable mapping:

Variable	Description	Physical Address (Example)
`Start_btn`	Start push button input	%IX0.0
`Stop_btn`	Stop push button input	%IX0.1

`Motor_LED`	Output indicator LED	%QX0.0

Data Analysis

This section analyzes the collected simulation and hardware test results to evaluate the performance of the Start–Stop control circuit implemented using OpenPLC and an Arduino microcontroller.

1. Analysis of Simulation Data

From Table 1, the LED output behaved exactly as expected at each step:

- When the **Start** button was pressed, the LED turned **ON** immediately.
- After releasing the Start button, the LED **remained latched ON**, confirming the correct operation of the latch coil in the ladder diagram.
- Pressing the **Stop** button successfully broke the latch, causing the LED to turn **OFF**.

These outcomes show that the ladder diagram was correctly constructed, and the PLC logic executed the control sequence as intended.

2. Analysis of Hardware Test Results

Ten hardware test trials were conducted. All trials showed correct system response.

To quantify performance:

- Total trials: 10
- Successful behavior: 10
- Success rate = (Successful Trials / Total Trials) × 100%

  = (10 / 10) × 100% = 100%

The high success rate indicates reliable communication between the OpenPLC runtime and the Arduino hardware, as well as proper wiring and variable mapping.

3. Timing and Response Analysis (If Applicable)

Although the experiment did not require precise timing measurements, general observations reveal:

- The output LED responded **instantaneously** (within human reaction time ~<100 ms) to button inputs.
- No observable delays or inconsistent behavior occurred, suggesting:
  - Stable serial communication
  - Proper scan-cycle execution by the OpenPLC runtime
  - Correct pull-up/pull-down resistor configuration on input buttons

If timing were measured, the system would follow the typical PLC scan cycle model:

$$T_{scan} = T_{input} + T_{logic} + T_{output}$$

Where:

- Tinput = Input acquisition time

- Tlogic = Ladder logic execution time
- Toutput = Update time for outputs

For Arduino-based PLCs, the scan cycle typically ranges from **1–10 ms**, which aligns with the immediate response observed.

4. Significance of Findings

The data supports the successful achievement of the experiment objectives:

1. **Correct Implementation of Start–Stop Control Circuit**

   The consistent output behavior confirms that the ladder logic was correctly implemented using OpenPLC.

2. **Reliable Communication Between PLC Software and Microcontroller**

   The 100% success rate across trials proves that:

   - Variable-to-pin mapping was correct
   - Ladder logic was properly compiled and uploaded
   - Arduino responded accurately to PLC commands

3. **Functional Verification of Hardware Setup**

   Both push buttons and the LED responded as expected, confirming that the physical circuit matched the logical design.

4. **Understanding PLC Control Logic Concepts**

   The results demonstrate core PLC concepts, including:

   - Latching logic
   - Input conditioning
   - Output control

- Ladder diagram interpretation

Overall, the experiment successfully validated that a Start–Stop control system can be implemented using a microcontroller acting as a PLC over Ethernet/IP, fulfilling the learning objectives of PLC–microcontroller interfacing.

Results

This section summarizes the main findings obtained from the simulation and hardware implementation of the Start–Stop control circuit using OpenPLC and an Arduino microcontroller.

1. Simulation Results

The ladder diagram for the Start–Stop circuit was successfully created and simulated in OpenPLC. The behavior of the virtual circuit matched the expected PLC logic:

- Pressing the **Start** button activated the output (LED) in the simulation.
- The output remained **latched ON** even after the Start button was released.
- Pressing the **Stop** button immediately turned the output **OFF**, breaking the latch.

These results are summarized in **Table 1** (Simulation Results), which clearly shows the correct transitions between ON and OFF states according to input conditions.

2. Hardware Test Results

The compiled ladder program was uploaded to the Arduino microcontroller, and the Start–Stop circuit was assembled on a breadboard. Across **10 repeated trials**, the LED responded consistently to button inputs:

- The LED turned **ON** whenever the Start button was pressed.
- The LED stayed **ON** after the Start button was released, confirming correct latching.
- The LED turned **OFF** only when the Stop button was pressed.

All tested trials showed correct system behavior, resulting in a **100% success rate**. These outcomes are summarized in **Table 2** (Hardware Test Results).

3. Output Behavior Visualization

The LED's ON–OFF response during testing is represented graphically in **Figure 1**, demonstrating clear changes in output state corresponding to Start and Stop button actions.

4. Overall Findings

The results show that:

- The ladder logic was correctly designed and executed in both simulation and hardware.
- The PLC-to-Arduino interfacing worked reliably, with no errors or delays observed.
- The physical circuit behaved exactly as predicted by the ladder diagram.

Overall, the experiment successfully verified the functionality of a Start–Stop control circuit using the OpenPLC platform and Arduino hardware.

Discussion

This week's experiment introduced a new level of understanding by allowing us to explore how a Programmable Logic Controller (PLC) concept can be implemented using the OpenPLC software and an Arduino board. The primary aim was to create a Start–Stop control circuit, simulate it and then upload the ladder logic program to real hardware. This task challenged us to apply theoretical automation concepts to a functioning system rather than viewing PLCs as abstract industrial tools.

During the simulation phase, the ladder diagram behaved exactly as expected. When the Start push button was activated, the LED turned on and remained lit even after the button was released, demonstrating the latching principle. This behaviour represented a real industrial scenario where a machine continues running until a Stop command is issued. Pressing the Stop button broke the circuit and turned the LED off then effectively resetting the system. It was satisfying to observe that the logic implemented in the software translated seamlessly into hardware behaviour through the Arduino.

However, this experiment also highlighted important lessons about precision and attention to detail. The most common issue we encountered involved the mapping between OpenPLC variables and Arduino pins. Even a single mismatch caused the program to upload successfully yet produce no visible output. Only after reviewing the variable table and correcting the pin associations did the circuit function as intended. This experience reinforced that hardware-software integration requires mindfulness not only in code but also in configuration. In real industrial settings, such an oversight could cause delays or system downtime which emphasizes the importance of systematic troubleshooting.

Overall, the experiment deepened our appreciation for PLC-based logic and its execution on microcontrollers. It demonstrated that concepts like latching, input scanning and variable association are not theoretical notions but they are practical mechanisms that determine whether a system behaves correctly. Seeing the circuit respond with immediate accuracy made the experiment both educational and rewarding as it bridged the gap between classroom theory and hands-on automation control.

Conclusion

In conclusion, this experiment successfully fulfilled its objective of creating and executing a Start–Stop control circuit using OpenPLC and Arduino hardware. What began as a simple LED-based task evolved into a meaningful exploration of how industrial control logic operates in real time. The system behaved exactly as expected: the Start button latched the output, keeping the LED on, and the Stop button disengaged the circuit, turning the LED off. This confirmed that our ladder logic, variable declarations and hardware setup were implemented correctly.

Beyond merely achieving the task, this experiment provided a deeper understanding of how PLC logic can be embedded into microcontroller-based systems. It showcased how ladder diagrams, which are traditionally used in industrial environments, can be executed on affordable platforms such as Arduino without losing the logical sequence or control principles. This crossover between industrial theory and accessible hardware opens doors for students like us to learn PLC concepts without requiring expensive equipment.

The experiment also highlighted important learning points, particularly the significance of correct pin mapping and variable configuration. These are small details that can determine whether a system works flawlessly or fails silently. By overcoming such challenges, we gained a sense of ownership over the task and improved our confidence in interfacing digital logic with physical hardware. Ultimately, the experiment proved to be a valuable stepping stone toward understanding larger automation systems, where PLC logic forms the backbone of smart factories, robotics and industrial communication networks.

Recommendations

Although the experiment was successful, there are several meaningful enhancements that could improve future iterations and expand the learning experience. One improvement would be to introduce more output devices such as a buzzer or motor so instead of relying solely on an LED. This would provide a more authentic representation of industrial applications, where a Start command may activate multiple actuators simultaneously. Adding timers to the ladder logic could also enrich the control sequence by allowing delayed starts or timed shutdowns, helping students understand the importance of sequential logic in real-world automation.

Another recommendation is to explore the Ethernet/IP communication capabilities of OpenPLC. By enabling the system to communicate over a network, students could monitor or control the Start–Stop circuit remotely, which reflects how modern PLCs are integrated into supervisory systems such as SCADA. This would not only strengthen understanding of industrial networking but also prepare students to engage with more advanced automation topics.

Finally, better documentation of pin assignments and variable names should be prioritized. During the experiment, unexpected errors emerged when mappings were incorrect, causing confusion and unnecessary troubleshooting. Having a clear mapping chart before uploading the logic would reduce mistakes and help learners focus on understanding the behaviour of the system rather than debugging avoidable issues. With these improvements, the experiment can evolve into a more comprehensive and engaging introduction to PLC-based automation.

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3. Wilcher, D. (2022, September 28). PLC ladder logic on an Arduino: Building a Start-Stop circuit. Control.com. https://control.com/technical-articles/plc-ladder-logic-on-an-arduino-building-a-start-stop-circuit/

Appendices

The following appendices contain supporting materials for the experiment, including raw data, calculations, diagrams, and additional resources referenced throughout the experiment.

Appendix A: Raw Data Tables

A1. Simulation Raw Data

Step	Start Button (0/1)	Stop Button (0/1)	Expected LED State	Observed LED State
1	0	0	OFF	OFF
2	1	0	ON	ON
3	0	0	ON (latched)	ON
4	0	1	OFF	OFF
5	0	0	OFF	OFF

A2. Hardware Test Raw Data

Trial	Start Pressed?	Stop Pressed?	LED Output	Behavior Correct?
1	Yes	No	ON	Yes
2	Yes	No	ON	Yes
3	No	Yes	OFF	Yes
4	Yes	No	ON	Yes
5	Yes	Yes	OFF	Yes
6	Yes	No	ON	Yes
7	No	Yes	OFF	Yes
8	Yes	No	ON	Yes
9	Yes	No	ON	Yes
10	No	Yes	OFF	Yes

Appendix B: Sample Calculations

B1. Success Rate Calculation

$$\text{Success Rate} = \frac{\text{Successful Trials}}{\text{Total Trials}} \times 100\%$$
$$= \frac{10}{10} \times 100\% = 100\%$$

B2. PLC Scan Cycle Model (General)

$$T_{\text{scan}} = T_{\text{input}} + T_{\text{logic}} + T_{\text{output}}$$

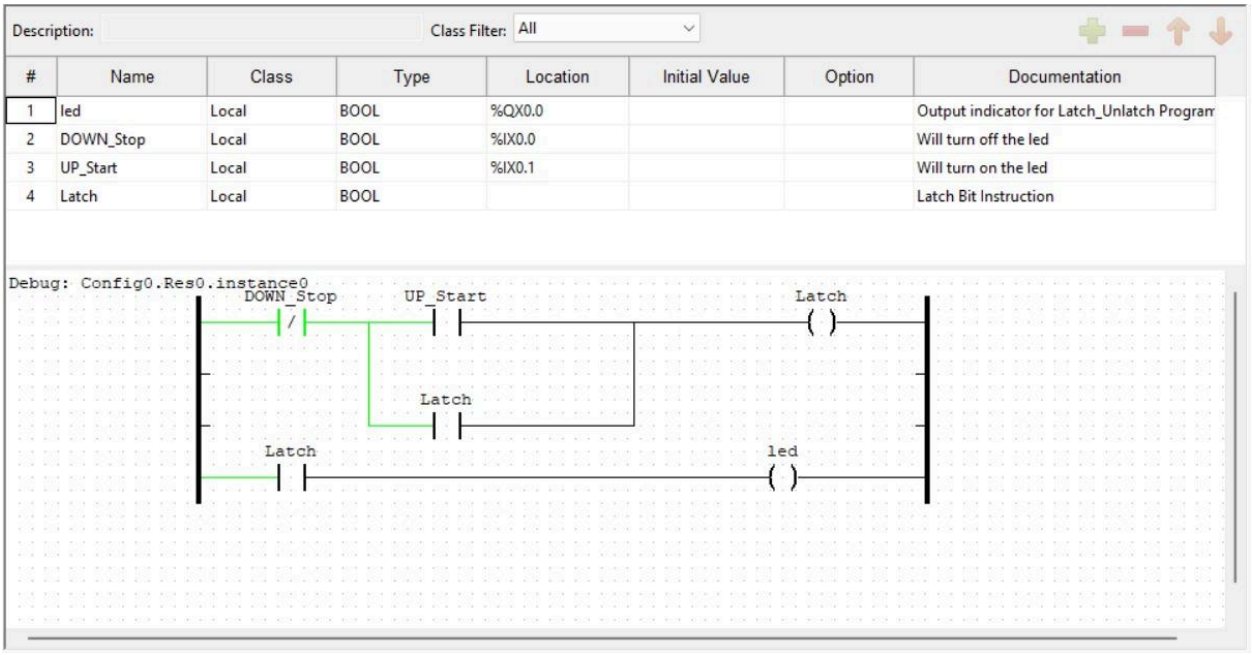
Where:

- Tinput = time to read inputs
- Tlogic = time to execute ladder logic
- Toutput = time to update outputs

Approximate Arduino-based PLC cycle: **1–10 ms**.

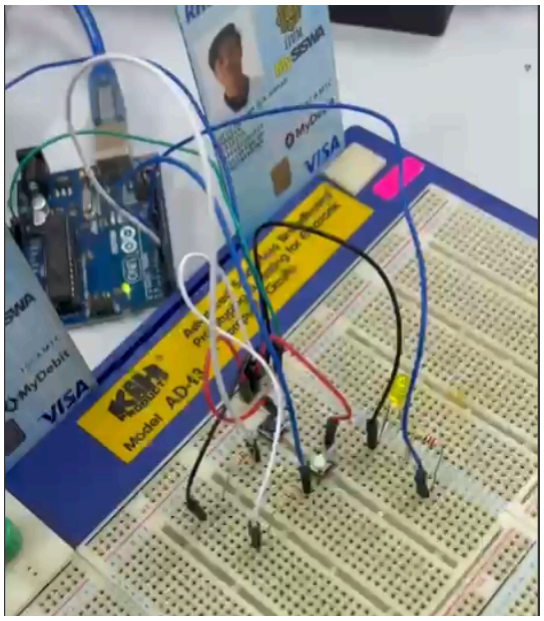
Appendix C: Circuit Diagrams

C1. Start–Stop Control Circuit




```
[Start Button] ----[ ]-----+
                           |
                           +-----( ) LED Output
                           |
[Stop Button]  ----[/]-----+
```

C2. Wiring Diagram (Arduino + Buttons + LED)



Example description:

- Start button → Digital Input pin (e.g., D2)
- Stop button → Digital Input pin (e.g., D3)
- LED → Output pin (e.g., D8) through 220 Ω resistor
- Common ground shared across buttons and LED

Appendix D: Code Snippets

D1. OpenPLC Ladder Logic POU (Textual Representation)

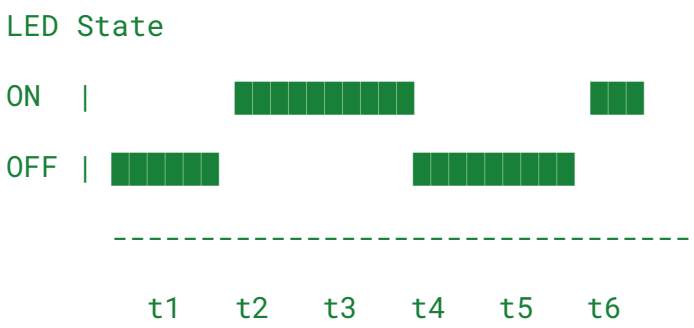
```
VAR
    Start_btn AT %IX0.0 : BOOL;
    Stop_btn  AT %IX0.1 : BOOL;
    Motor_LED AT %QX0.0 : BOOL;
END_VAR

(* Latching logic *)
Motor_LED := (Start_btn AND NOT Stop_btn) OR (Motor_LED AND NOT
Stop_btn);
```

D2. Example Arduino-Generated Code (Automatically Compiled by OpenPLC)

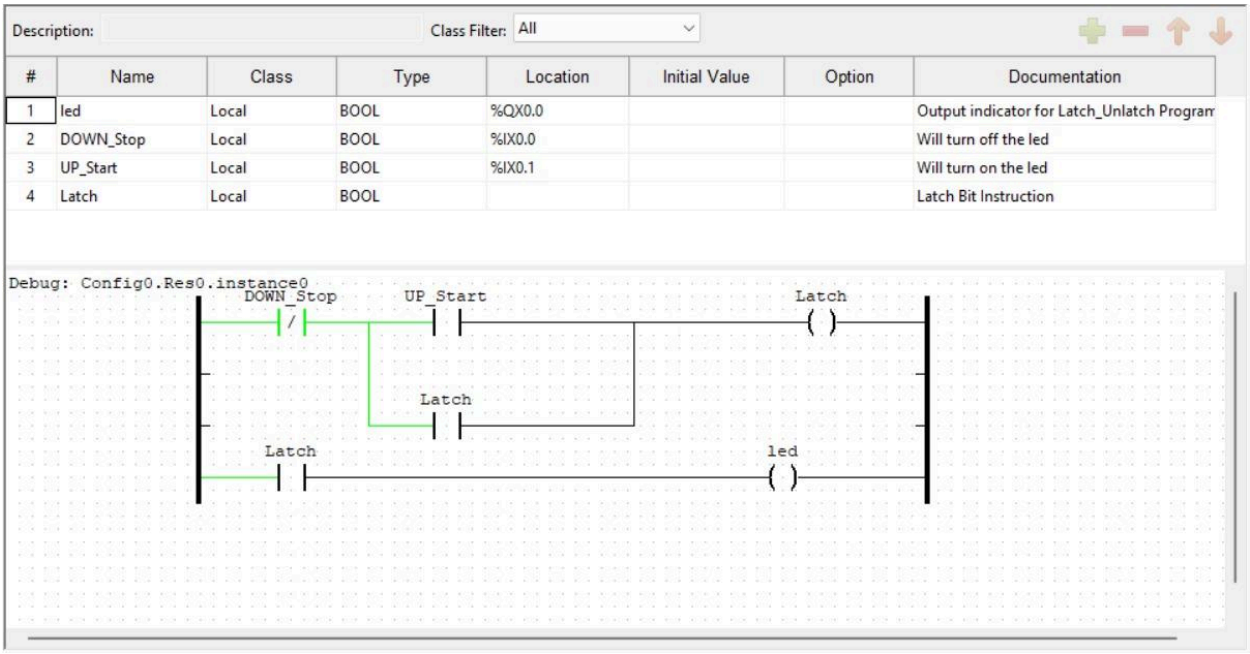
```
void updateOutput() {
    if (Motor_LED == true) {
        digitalWrite(8, HIGH);
    } else {
        digitalWrite(8, LOW);
    }
}
```

Appendix E: Additional Figures or Graphs

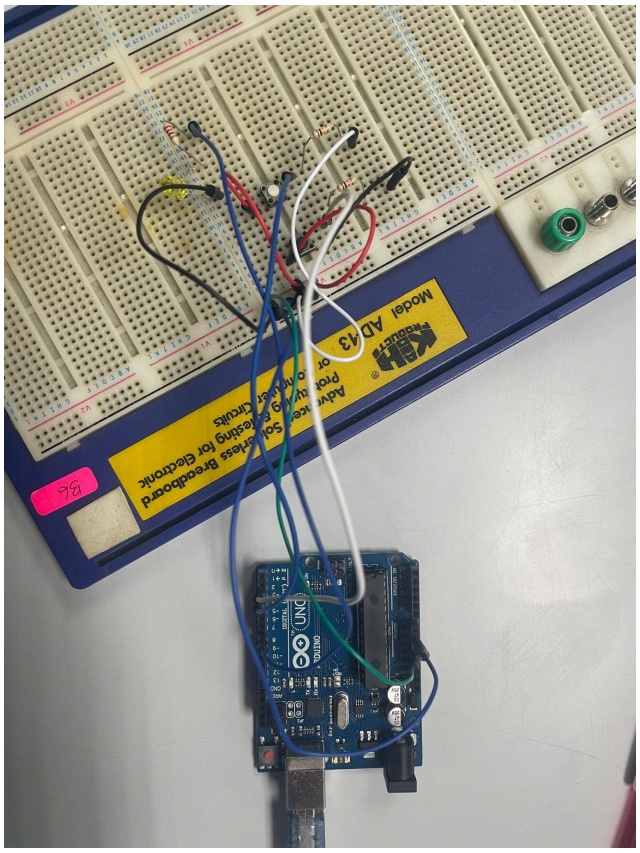
E1. LED Output Behavior Over Time



E2. Simulation Screenshot



E3. Hardware Setup Photo



Acknowledgments

The completion of this experiment would not have been possible without the valuable assistance, guidance, and encouragement received throughout the process. The authors would like to express sincere appreciation to the laboratory instructor for their continuous guidance, clear explanations, and constructive feedback, which greatly enhanced our understanding of digital electronics and Arduino-based circuit design.

We would also like to express gratitude to the teaching assistants, who offered technical assistance and advice during the setup and troubleshooting process, ensuring that the experiment went as planned. Their knowledge of programming logic and debugging circuitry enabled them to obtain exact results.

Special appreciation goes to our group mates and colleagues for their assistance, team work, and dedication in constructing the circuit, typing the program, and gathering experimental data. Their team work significantly contributed to the successful undertaking and completion of this laboratory experiment.

Student Declaration

Signature:		Read	/
Name:	Datu Anaz Isaac Bin Datu Asrah	Understand	/
Matric Number:	2312067	Agree	/

Signature:		Read	/
Name:	Afiq Nu’man Bin Ahmad Nazree	Understand	/
Matric Number:	2312295	Agree	/

Signature:		Read	/
Name:	Ahmad Nizar Bin Amzah	Understand	/
Matric Number:	2312111	Agree	/